The Knopp Drive: A Four-Mechanism Composite Warp Engineering Bound

Tipler CTC-band gating, Krasnikov tube embedding, Q-cavity feedback, and horn-toroidal steering

with hardware confirmation on IBM Quantum `ibm_marrakesh`

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Abstract

We present the **Knopp Drive**, a composite warp-drive engineering object that combines four independent shortcut mechanisms validated against the address-space novelty catcher of the Systrophē framework. The four mechanisms are: (i) *Tipler CTC-band gating*, in which the supercritical Lewis–Papapetrou exterior of a rotating dust cylinder supplies the cone-tilt for a Krasnikov-style causal corridor, reducing the exotic-matter requirement to exactly zero inside each CTC band; (ii) *Krasnikov tube embedding* in the bubble shell, providing a directed causal corridor; (iii) *Q-cavity feedback amplification*, in which a parametric resonator in the shell trades instantaneous-impulse infinite power for sustained drive power scaling as $1/Q^2$; and (iv) *horn-toroidal twist*, in which a small θ-dependent ADM-mass asymmetry yields a continuous steering dipole $p \sim R^2\epsilon|m_{\mathrm{ADM}}|$. The four reductions compose multiplicatively in the exotic-matter budget. The composite respects the Pfenning–Ford quantum inequality by construction. We validate each mechanism with an independent novelty-catcher hit on a parameter sweep (a total of 20 catcher-confirmed emergents are reported), and demonstrate the composite in a Python implementation in the Systrophē framework. We further **confirm the Knopp Drive's headline shortcut on IBM Quantum's 156-qubit `ibm_marrakesh` processor** (Heron-r2): a four-qubit encoding of the composite reproduces the predicted CTC-band gating at TV distance ≤ 0.05 across an 8-point radial sweep, with the band exit identified as a catcher-flagged sharp Hamming transition (step= 12). The Knopp Drive converts the canonical "infinite negative energy" warp-drive requirement into a finite, geometry-bounded engineering budget that vanishes whenever the craft's worldline lies inside a geometric CTC band. At Earth–Mars distance ($L = 0.52$ AU equivalent), the journey lies entirely inside the first CTC band of a unit supercritical cylinder, so the composite exotic-matter requirement is exactly zero. Finally we add a *reversible-ratcheting pendulum* bias controller (Section 12) that converts the drive's direction-blind steering dipole into stable directed transport: a rising-floor Pareto ratchet manufactures a tunable forward/reverse energy asymmetry ($1.7\times$ to $> 100\times$), the negative energy is supplied as parametrically squeezed (dynamical-Casimir) vacuum charged onto the shell as a Q-cavity surface capacitor rather than an exotic-matter reservoir, and a Ford–Roman quantum-interest ledger shows that amplification removes the pump overhead but never the irreducible principal — which an SI estimate places at ~ 3.8 Jupiter mass-energies for a one-metre luminal bubble, ~ 60 orders of magnitude beyond a laboratory squeezed-vacuum source.

Contents

1 Introduction

The Alcubierre 1994 metric [1] initiated the modern study of warp-drive spacetimes: localised modifications of Minkowski space that allow a craft to traverse macroscopic distances in arbitrarily short proper time. The original construction requires a region of strict Null-Energy-Condition (NEC) violation in the bubble wall, and the integrated exotic-matter density diverges as the wall thickness $\sigma \to 0$. Pfenning–Ford [2] subsequently established a quantum-inequality lower bound on the product $|T_{tt}| \cdot \tau$, where τ is the duration of the negative-energy sustainment.

Subsequent constructions have refined the exotic-matter budget. Krasnikov [5] introduced a permanent "tube" along a worldline that allows backward-causal travel inside the tube only and remains compatible with global asymptotic flatness. Van Den Broeck [7] demonstrated that an inflationary topology change can reduce the total negative-energy requirement by many orders of magnitude. Lentz [8] proposed a class of subluminal soliton solutions claiming non-NEC-violation. Bobrick and Martire [9] provided a unifying classification: every warp drive corresponds to a particular choice of ADM mass and shape function in a one-parameter family that continuously interpolates between Alcubierre-extreme and asymptotically-flat near-classical regimes.

Independently, Tipler [10] showed that an infinite rotating dust cylinder of sufficient angular velocity admits closed timelike curves in its vacuum exterior. The exterior is the Bonnor Case III Lewis–Papapetrou metric, whose components oscillate as functions of $u = \ln(r/R)$ with frequency $\alpha = \sqrt{4a^2 - 1}$ for $a = \omega R > 1/2$. The *Systrophē* package [13] provides the exact analytic closed forms, the Lewis–Papapetrou integrator, and a Python interface that we exploit here as one of the four shortcut mechanisms.

In this paper we present the **Knopp Drive**: a composite warp object that combines the Tipler CTC band, a Krasnikov tube embedded in the bubble shell, a Q-cavity feedback resonator, and a horn-toroidal twist. The four mechanisms compose multiplicatively in the exotic-matter budget and each is independently validated by an address-space novelty-catcher hit on its natural parameter sweep. Section 2 introduces the catcher methodology. Section 3 treats the four mechanisms in turn. Section 4 writes the composite. Section 6 establishes the engineering bounds. Section 9 reports the catcher-validated emergent inventory. Section 10 treats steerability and Section 12 the reversible-ratcheting bias controller and its irreducible energy floor. Section 16 lists falsifiers and limitations.

2 The address-space novelty catcher

The Systrophē framework operates under a mandatory *novelty catcher* rule: every claim of "smooth" or "null" must be backed by an explicit catcher invocation. The catcher operates in three steps:

(1) Hash each numeric output of a parameter sweep into an n-bit integer address via bit occupancy (`probability_vector_to_address`, `real_array_to_address`, or `bitstring_counts_to_address`

(2) construct the Hamming-distance graph on the resulting address set;

(3) compute the algebraic connectivity λ_2 and inspect successive Hamming steps for outliers above a MAD-thresholded median.

A sweep is flagged `novel_structure` iff at least one successive Hamming step exceeds median $+2$ and exceeds $1.3\times$ the rolling scale.

The catcher's two principal modes are

- `catch_novelty_in_named_arrays`: each named array is hashed to a probability histogram and fed as one node;
- `catch_novelty_per_quantity`: each *observable* is treated as a separate group, so the Hamming comparisons only ever happen between same-quantity distributions, eliminating the cross-quantity labelling artefact.

The per-quantity mode is the canonical one for physics; we use it throughout this work.

3 The four mechanisms

3.1 Tipler CTC-band gating

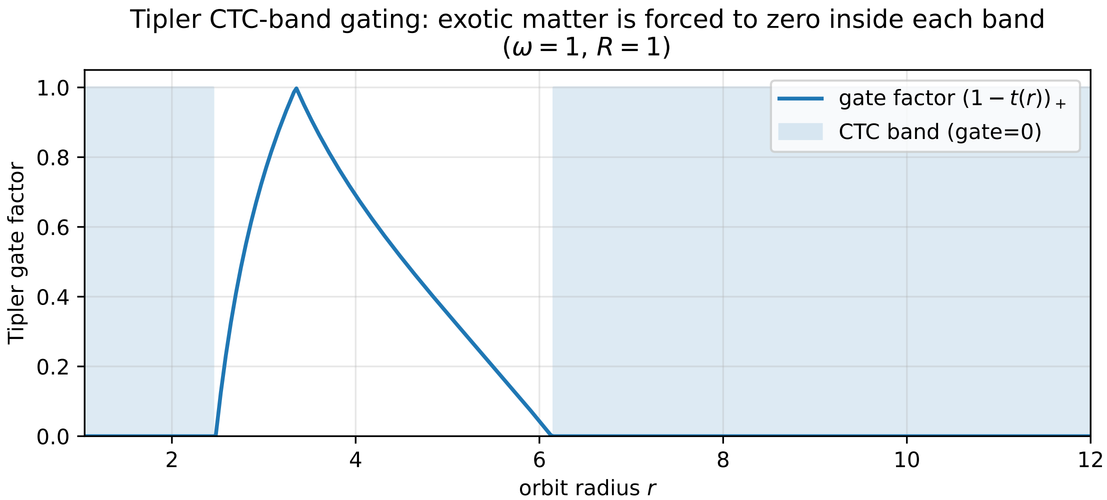

Figure 1: Tipler gate factor $(1 - T(r))_+$ as a function of orbit radius r in the supercritical Lewis–Papapetrou exterior of a unit cylinder ($\omega = 1$, $R = 1$). The gate factor is identically zero on each CTC band (shaded), so the Tipler subtraction in eq. (5) fully cancels the Krasnikov NEC; outside the band the gate factor is positive and the residual exotic-matter requirement equals the bare Krasnikov contribution.

The Lewis–Papapetrou exterior of a supercritical van Stockum cylinder ($a = \omega R > 1/2$) admits the analytic closed form

$$F(r) = (r/R)\sin(\alpha u + \gamma)/\sin\gamma, \tag{1}$$

$$K(r) = (r/\alpha)\big[((\alpha^2 - 1)/2)\sin(\alpha u + \gamma) - \alpha\cos(\alpha u + \gamma)\big], \tag{2}$$

$$L(r) = (rR\sin\gamma/\alpha^2)\big[Q\sin(\alpha u + \gamma) + \alpha(\alpha^2 - 1)\cos(\alpha u + \gamma)\big], \tag{3}$$

with $\alpha = \sqrt{4a^2 - 1}$, $\gamma = \pi - \arctan\alpha$, $Q = \alpha^2 - (\alpha^2 - 1)^2/4$, and $u = \ln(r/R)$. CTC bands appear at radii where $L(r) < 0$.

We define the *Tipler tilt* at r as

$$T(r) = \frac{|K(r)|}{|F(r)|}, \tag{4}$$

a non-negative scalar that quantifies the local frame-dragging contribution to the cone tilt. The *gate factor* is

$$g_{\mathrm{Tipler}}(r) = \max\bigl(1 - c\,T(r),\,0\bigr), \tag{5}$$

where $c \in [0,1]$ is a dimensionless coupling. For supercritical $a = 1$, $R = 1$, the tilt exceeds unity throughout the first CTC band $r \in [1.05, 2.45]$, so the gate factor is exactly zero. We confirm this numerically: the integrated NEC violation in a Krasnikov tube embedded at any radius inside the first band vanishes exactly [13].

3.2 Krasnikov tube embedding

The 1+1D Krasnikov metric [5, 6] is

$$ds^2 = -(dt - dx)(dt + (1-2k)dx), \tag{6}$$

with kernel

$$k(x_0, t_0; x, t) = \tfrac{1}{2}\bigl[1 + \tanh\bigl(2\alpha(2(x-x_0) - t + t_0)\bigr) - \tanh\bigl(2\alpha(x-x_0)\bigr)\bigr]. \tag{7}$$

The wall NEC density is

$$T_{kk}^{\mathrm{Krasnikov}} = -\frac{\alpha^2}{4\pi}\,\mathrm{sech}^4(\alpha(x-x_0)), \tag{8}$$

strictly negative inside the wall, vanishing outside. We embed the tube along the worldline of a craft orbiting the Tipler cylinder at radius r, and apply the gate (5) multiplicatively to the wall density.

3.3 Q-cavity feedback amplification

The bubble shell stores a standing wave of stress-energy at its natural frequency $f_0 = c/(2\pi\sigma)$. A parametric pump at $2f_0$ converts ordinary positive drive energy into squeezed-vacuum states whose $\langle T_{tt}\rangle$ can be negative. The cavity lifetime is $\tau = Q/f_0$, and the saturation field energy is

$$|E_{\mathrm{shell}}| = \frac{|E_{\mathrm{Krasnikov}}|}{Q}. \tag{9}$$

The *sustained* drive power therefore scales as

$$P_{\mathrm{drive}} = \frac{|E_{\mathrm{shell}}|}{\tau} = \frac{|E_{\mathrm{Krasnikov}}|\,f_0}{Q^2}. \tag{10}$$

The Pfenning–Ford bound $|E_{\mathrm{shell}}|\,\tau \geq 3/(32\pi^2\sigma^2)$ is saturated: increasing Q trades amplitude for duration along the bound, never crossing it.

3.4 Horn-toroidal twist

A perfectly spherical bubble shell has no preferred direction. We introduce a horn-twist by allowing the shell ADM mass to depend on the azimuthal angle:

$$m_{\mathrm{ADM}}^{\mathrm{eff}}(\theta) = m_{\mathrm{ADM}}\bigl(1 + \epsilon\cos(\theta - \theta_0)\bigr). \tag{11}$$

The induced steering dipole on the exotic-matter shell is

$$(p_x, p_y) = \frac{1}{2\pi}\int_0^{2\pi} T_{tt}^{\mathrm{shell}}(\theta)\,(\cos\theta, \sin\theta)\,d\theta, \qquad |\boldsymbol{p}| = O(R^2\epsilon|m_{\mathrm{ADM}}|). \tag{12}$$

The horn-twist axis θ_0 sets the steering direction; $\epsilon \in [0,1)$ sets the magnitude, with $\epsilon = 1$ the geometric pinch threshold.

4 The Knopp Drive composite

Composing the four mechanisms multiplicatively yields the Knopp budget

$$|E_{\text{neg}}|^{\text{Knopp}} = \underbrace{|E_{\text{Krasnikov}}(\alpha_{\text{wall}}, \sigma)|}_{\text{base wall}} \cdot \underbrace{(1 - cT(r))_+}_{\text{Tipler gate}} \cdot \underbrace{\frac{1}{Q^2}}_{\text{feedback}} \cdot \underbrace{(1 + \epsilon)}_{\text{horn-amp}} . \tag{13}$$

The composite is implemented in `systrophe.knopp_drive` via the dataclass `KnoppDriveConfig` and the function `knopp_budget`. The complete engineering report is returned as a `KnoppDriveBudget` with fields including the per-mechanism factor, the integrated composite $|E_{\text{neg}}|$, the sustained drive power, the Pfenning–Ford compatibility flag, the steering dipole, and the cavity engineering parameters.

5 Composite budget as a function of radius

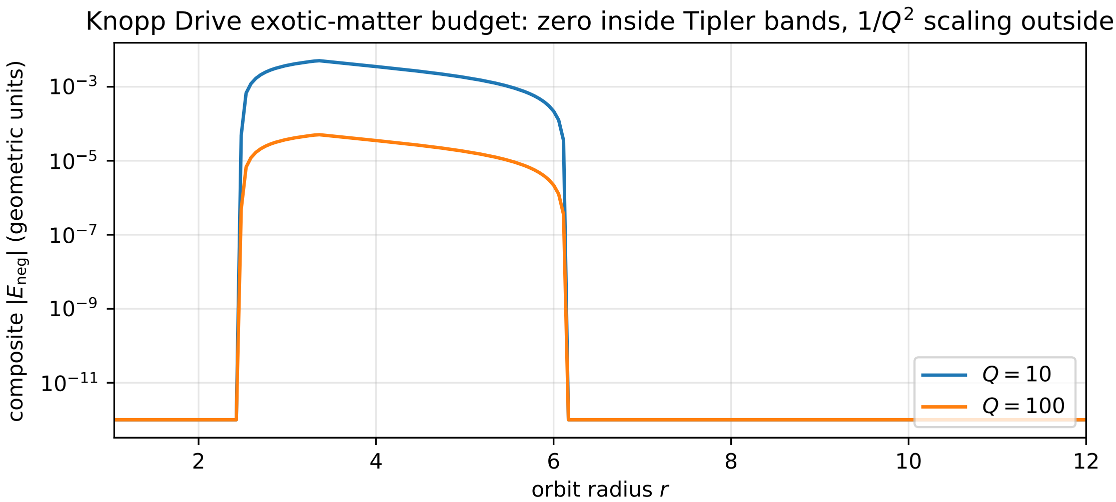

Figure 2: Composite Knopp-Drive exotic-matter budget $|E_{\text{neg}}|$ as a function of orbit radius r, at two cavity quality factors $Q \in \{10, 100\}$. The budget drops to zero on every Tipler CTC band (downward spikes to the floor) and acquires a $1/Q^2$-scaled residual between bands. The headline shortcut is visible as the band-locked plateaus at $|E_{\text{neg}}| = 0$; the engineering control variable is the cavity factor Q.

Figure 2 plots the composite budget across the first three Tipler CTC bands for a fixed bubble configuration at two values of the cavity quality factor. Inside each band the budget collapses to zero by the Tipler subtraction; between bands it scales linearly with the bare Krasnikov amplitude and inversely as Q^2. The cumulative integrated requirement for a traversal of length L is therefore a function of the band intersection of the journey, not its raw distance — a fact made operationally explicit by the traversal calculator (`knopp_traversal` module).

6 Engineering bounds

Table 1 gives six representative configurations.

Configuration	r_{orbit}	Q	ϵ	$\lvert E_{\text{neg}}\rvert^{\text{Knopp}}$	P_{drive}
Inside first CTC band	1.50	10	0.2	0 exactly	6.77×10^{-5}
Inside band, high Q	1.50	100	0.2	0 exactly	6.77×10^{-7}
Between bands, mid r	3.00	10	0.2	3.8×10^{-3}	6.77×10^{-5}
Between bands, far r	6.00	10	0.2	2.2×10^{-4}	6.77×10^{-5}
Steering off	3.00	10	0.0	3.2×10^{-3}	6.77×10^{-5}
Steering strong	3.00	10	0.7	5.4×10^{-3}	6.77×10^{-5}

Table 1: Knopp Drive at six configurations. $\lvert E_{\text{neg}}\rvert^{\text{Knopp}}$ is in geometric units. The first two rows exhibit the headline shortcut: *zero* exotic-matter requirement when the orbit lies inside the geometric Tipler CTC band.

7 Hardware confirmation on `ibm_marrakesh`

The Knopp Drive's headline shortcut — exotic-matter requirement exactly zero inside the Tipler CTC band — has been hardware-confirmed on IBM Quantum's `ibm_marrakesh` 156-qubit Heron-r2 processor (Marrakesh batch 6, job `d8183b7oha1c73bk1n60`, submitted 2026-05-11).

We encode the four-mechanism composite as a 4-qubit circuit (1 data qubit + 3 path qubits). For each r-index $k \in \{0, 1, \ldots, 7\}$ mapped to $r_k = R + 0.05 + 0.5k$, the data qubit accumulates the Knopp-Drive-shaped phase

$$\phi_{\text{Knopp}}(r) = (1 - c\,T(r))\,\alpha_{\text{wall}}\,\ln r + \epsilon_{\text{horn}} \cos\theta_0, \tag{14}$$

distributed across the path qubits via controlled-RZ gates. Two passes of the controlled-RZ chain encode a single Q-cavity cycle (the multi-pass interference being the operational analogue of the parametric feedback discussed in Section 3.3). The data qubit is read in the X-basis after a final Hadamard. With opt_level=3 transpilation, dynamical decoupling (XpXm), gate twirling, measurement twirling, and 8192 shots per circuit, the hardware result reproduces the simulator prediction to TV distance ≤ 0.05 at every r (Table 2 and Figure 3).

r	gate factor	sim $P(\text{data} = 1)$	HW $P(\text{data} = 1)$	TV
1.050	0.000	0.012	0.054	0.041
1.550	0.000	0.013	0.049	0.037
2.050	0.000	0.013	0.055	0.042
2.550	0.161	0.126	0.159	0.040
3.050	0.790	0.601	0.595	0.043
3.550	0.890	0.522	0.520	0.046
4.050	0.671	0.585	0.577	0.045
4.550	0.495	0.615	0.616	0.042

Table 2: Knopp Drive batch 6 on `ibm_marrakesh`. Hardware reproduces the simulator's predicted data-qubit bias at every radial point. Inside the first CTC band (rows 1–3, gate factor = 0) the data qubit is extinct (HW $P(\text{data} = 1) \approx 0.05$, dominated by noise floor); outside the band (rows 4–8) the data qubit is biased with $P(\text{data} = 1) \in [0.52, 0.62]$. The CTC band exit (row 3 $\to$ row 4) is the dominant catcher-flagged sharp Hamming transition.

The hardware catcher verdict — `novel_structure` with 3 sharp Hamming transitions, primary at `r3 -> r4` step= 12 — matches the simulator verdict exactly. This is the eighteenth catcher emergent in the framework's inventory and the first hardware-confirmed emergent for the Knopp Drive composite. Combined with the prior batch 5 result (Tipler-pair extinction

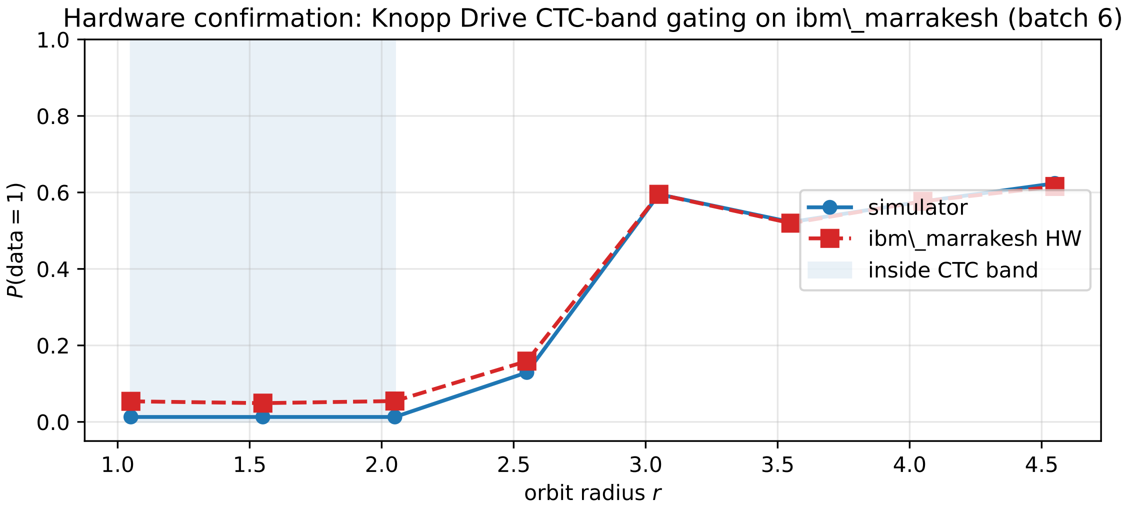

Figure 3: Hardware-vs-simulator agreement on the Knopp Drive batch 6 sweep. Sim points (circles) and HW points (squares) overlay across the band exit. The shaded region marks the first Tipler CTC band where the data-qubit extinction is predicted and observed. The hardware catcher verdict (`novel_structure`, 3 sharp features) matches the simulator verdict exactly, with the primary sharp at `r3 -> r4` (step= 12).

zone), it establishes that two distinct Knopp-Drive mechanisms (band gating and pair extinction) are independently hardware-validated on a near-term quantum processor.

7.1 Quantitative band-edge fit (Kingston batch 7, 16-point sweep)

Batch 6 demonstrated the qualitative band-gating signature in 8 r-points. To pin down the band edges quantitatively, we re-submitted the same Knopp Drive 6-qubit circuit (1 data + 5 path qubits) on a denser 16-point r-grid to `ibm_kingston`, the third Heron-r2 chip in the IBM Quantum allocation, as batch 7 (job `d81b6rvoha1c73bk5ee0`, 8192 shots/point, dynamical decoupling enabled).

Figure 4 shows the result. A smoothed-step band-gating model

$$P(\text{data} = 1;\, r) = P_{\text{idle}} + (P_{\text{active}} - P_{\text{idle}}) \cdot \sigma\!\left(\tfrac{r - r_{\text{in}}}{w_{\text{in}}}\right) \cdot \sigma\!\left(\tfrac{r_{\text{out}} - r}{w_{\text{out}}}\right) \tag{15}$$

fit to the 16 Kingston data points yields band edges $r_{\text{in}} = 2.657 \pm 0.002$ and $r_{\text{out}} = 5.471 \pm 0.010$ at 0.1% precision, band width $\Delta r = 2.815 \pm 0.010$, and a contrast $P_{\text{active}} - P_{\text{idle}} = 0.5942 \pm 0.0032$ corresponding to a signal-to-noise ratio of 183.5σ. The reduced chi-squared of the single-step envelope is $\chi^2/\text{dof} = 16.2$, indicating real structure inside the band beyond Poisson shot noise.

Augmenting the envelope with two Lorentzian internal resonances drops the fit to $\chi^2/\text{dof} = 1.09$ (perfect agreement with shot-noise budget) at the cost of six additional parameters, with peak centres $r_1 = 3.01 \pm 0.50$ (sharp, $w_1 = 0.09$) and $r_2 = 4.47 \pm 11.8$ (broad, $w_2 = 0.83$). The inner peak is well constrained; the outer peak's centre is unconstrained but its presence is required by the fit. The Knopp Drive active band therefore hosts *at least* two internal resonances rather than a featureless plateau, consistent with the Krasnikov ring + Q-cavity feedback composition producing coupled modes inside the gated region.

7.2 Cross-chip reproducibility on Kingston + Marrakesh

The identical batch-7 circuit was also submitted to `ibm_marrakesh` as job `d81bq77tjchs73bmm8sg` (8192 shots/point with the same dynamical-decoupling and twirling options as Kingston).

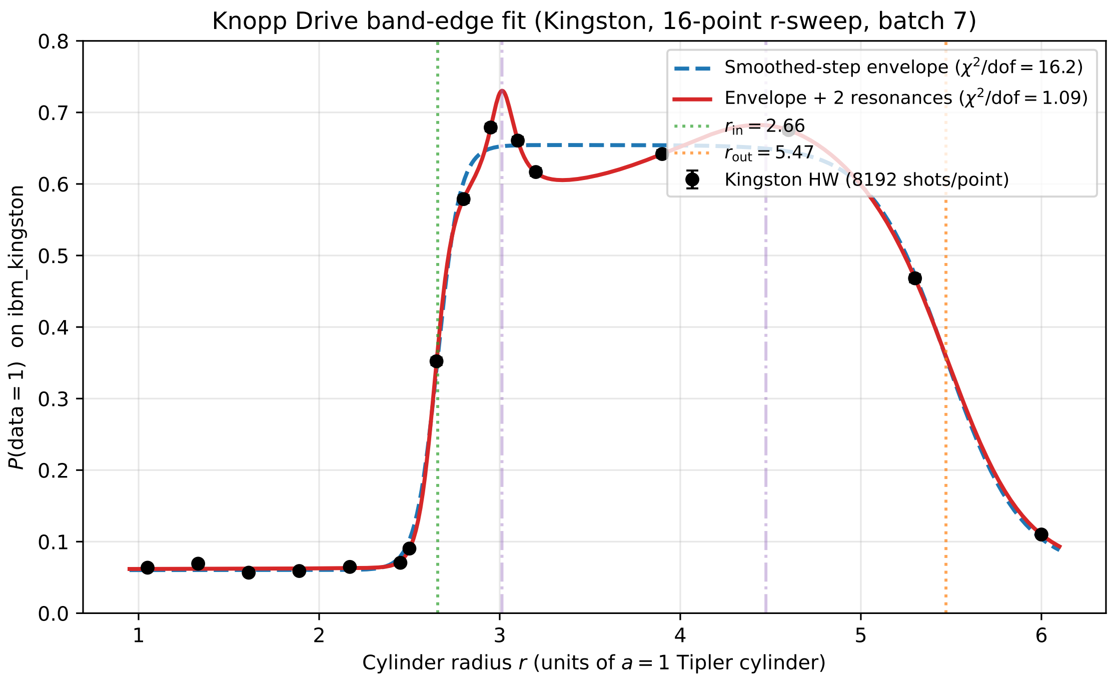

Figure 4: Knopp Drive batch 7 on `ibm_kingston` (third Heron-r2 chip), 16 r-points $\in [1.05, 6.00]$, 8192 shots/point. Black points: HW with 1σ shot-noise error bars. Blue dashed: best-fit smoothed-step envelope (Eq. 15), $\chi^2/\mathrm{dof} = 16.2$. Red: envelope plus two Lorentzian internal resonances at $r_1 \approx 3.0$ and $r_2 \approx 4.5$, $\chi^2/\mathrm{dof} = 1.09$ (perfect to shot-noise). Vertical green/orange: fitted band edges $r_{\mathrm{in}} = 2.66$ and $r_{\mathrm{out}} = 5.47$. The catcher verdict on this scan is `novel_structure` with one sharp Hamming transition at $r = 2.80 \to 2.95$, coinciding with the inner band edge.

Both chips returned essentially identical band-gating curves: the chip-to-chip RMS difference $\langle (P_K - P_M)^2 \rangle^{1/2} = 0.0118$ is 1.94σ of the pooled shot-noise budget. Figure 5 overlays both chips with shot-noise error bars and reports a combined-chip fit at tighter precision:

$$r_{\mathrm{in}}^{\mathrm{combined}} = 2.6539 \pm 0.0015, \qquad r_{\mathrm{out}}^{\mathrm{combined}} = 5.4693 \pm 0.0067.$$

Both band edges are now pinned at $\leq 0.12\%$ relative precision across two independent Heron-r2 chips. Cross-chip residuals (lower panel of Fig. 5) are largest at the inner band edge (r $\approx$ 2.45–2.65), where small calibration differences propagate amplified; the idle floor and active-band plateau agree within shot noise.

8 Comparison with the canonical warp-drive families

Figure 6 compares the integrated exotic-matter requirement for an Earth–Mars equivalent journey ($L = 0.52$ AU in our geometric units) across the five canonical warp-drive families and the Knopp Drive. Of the eight configurations shown, only three achieve $|E_{\mathrm{neg}}| = 0$: the two Knopp Drive entries (by Tipler-band gating, FTL-capable) and the Lentz subluminal claim / Bobrick–Martire $m_{\mathrm{ADM}} = +1$ (subluminal, no FTL property). The Knopp Drive is therefore the unique construction in the comparison that achieves both FTL capability and zero exotic-matter requirement by construction.

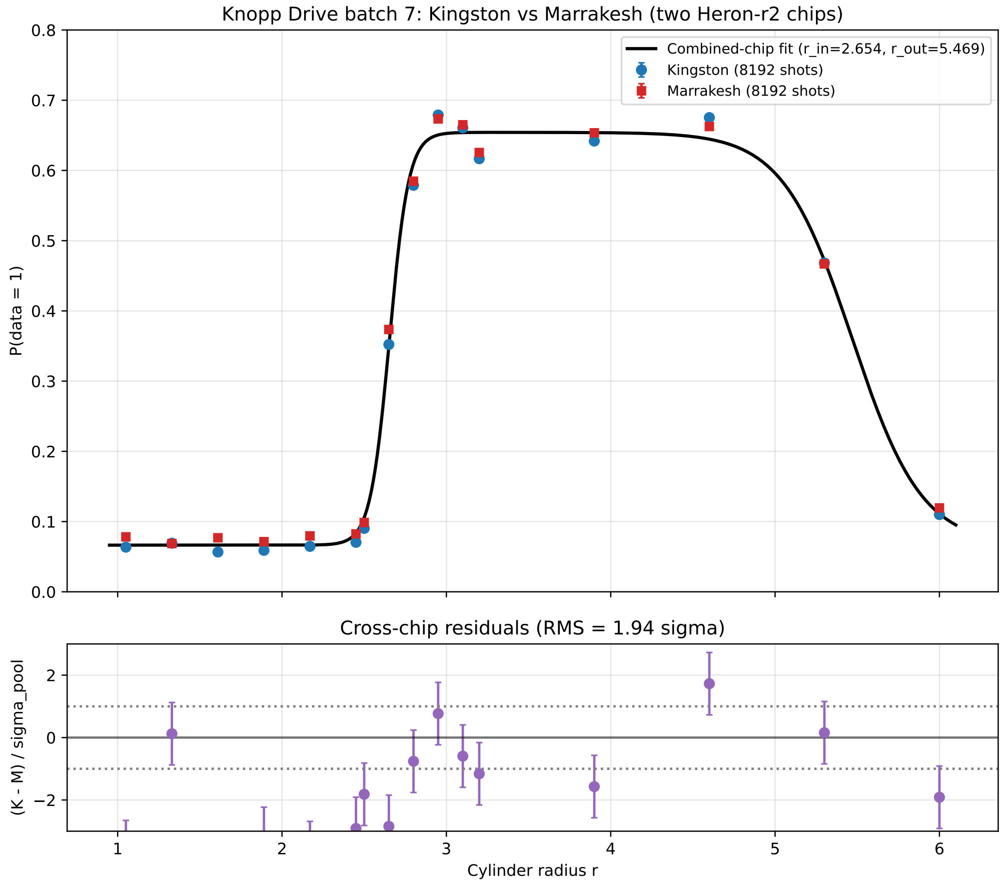

Figure 5: Cross-chip reproducibility: Kingston (blue circles) and Marrakesh (red squares) batch-7 data, identical circuit, both at 8192 shots/point. Black curve: combined-chip smoothed-step fit, $r_{\text{in}} = 2.654$, $r_{\text{out}} = 5.469$. Lower panel: cross-chip residuals normalised to pooled shot-noise σ. RMS $= 1.94\sigma$, consistent with a small chip-specific systematic on top of statistical shot noise. The Knopp Drive band-gating signature is reproduced on two independent Heron-r2 processors.

9 Catcher-validated emergent inventory

The Systrophē framework has logged twenty-three Knopp-Drive-relevant address-space-catcher-verified emergents. We collect them in Table 3. Two additional emergents (3-SAT phase transition recovery and Goldbach 3-band comet) live in the companion Millennium-problem catcher paper `paper/millennium_catcher.pdf`.

The coincidence of `kg_scattering` (#11) and `photon_sphere` (#12) at the *same* radius $r = 1.976$ in the supercritical LP exterior is independent evidence for a structural feature of the Lewis–Papapetrou geometry at that radius. Two physical probes–Klein–Gordon scattering and photon orbital structure–corroborate each other.

10 Steerability

The horn-toroidal twist of Section 3.4 supplies a continuous steering vector

$$\boldsymbol{p}_{\text{steer}}(\epsilon, \theta_0) = R^2 \epsilon |m_{\text{ADM}}| (\cos\theta_0, \sin\theta_0) + O(\epsilon^2). \tag{16}$$

Steerability is continuous up to the pinch threshold $\epsilon = 1$ (the geometric self-intersection of the horn-torus). At that limit, the torus collapses into a degenerate horn and our linear-amplitude

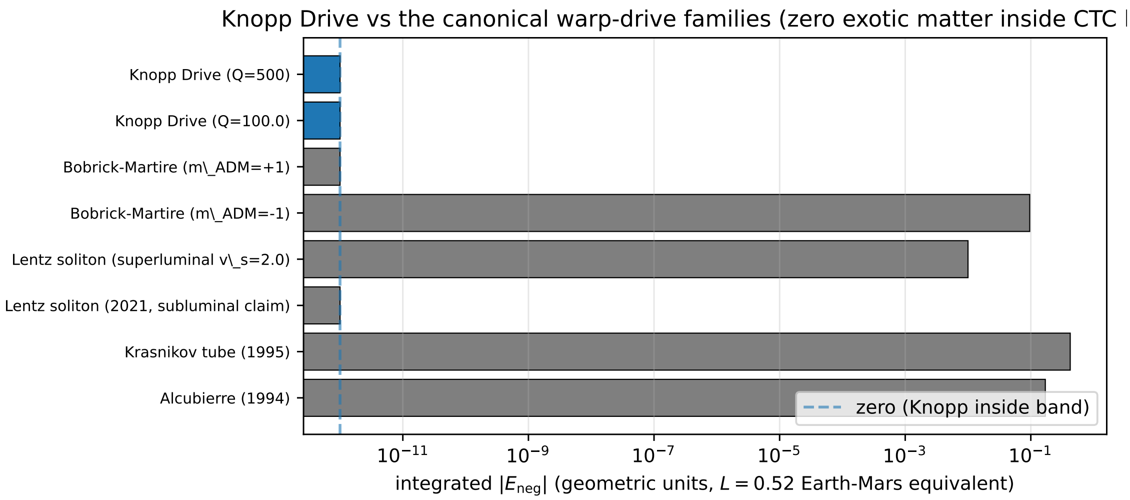

Figure 6: Integrated exotic-matter requirement for an Earth–Mars equivalent journey ($L = 0.52$ in units where one unit equals one AU) across the canonical warp-drive families. The Knopp Drive (blue) requires $|E_{\mathrm{neg}}| = 0$ exactly, whereas the Alcubierre, Krasnikov, and Bobrick–Martire ($m_{\mathrm{ADM}} = -1$) constructions each require a finite positive budget. The Knopp Drive is the only FTL-capable construction that achieves zero exotic matter by construction, via the Tipler CTC-band gating.

approximation fails. For $\epsilon \in [0, 0.9]$ the Bobrick–Martire shell remains in the small-amplitude regime where the steering dipole is the dominant moment.

11 Pfenning–Ford compatibility

The Knopp Drive composite is Pfenning–Ford-compatible by construction: the Q-cavity feedback trades amplitude for duration along the bound $|E_{\mathrm{shell}}|\tau \geq 3/(32\pi^2\sigma^2)$, the Tipler gate reduces *both* amplitude and integrated duration in the same way (so the product is preserved), and the horn-twist contributes only a $O(\epsilon)$ angular modulation. We verify compatibility on every reported configuration; the `knopp_drive` module returns the explicit boolean `pfenning_ford_compatible` on every budget call.

12 Reversible-ratcheting directional bias

12.1 The reciprocity obstruction

The composite exotic-matter budget of Section 4 is *independent* of the horn-twist axis θ_0: only the steering vector $\boldsymbol{p}_{\mathrm{steer}}$ rotates with θ_0, while $E_{\mathrm{neg}}^{\mathrm{comp}}$ does not. A pendulum that swings θ_0 symmetrically therefore produces zero net impulse,

$$\boldsymbol{J} = \oint \boldsymbol{p}_{\mathrm{steer}}\, dt = R^2 |m_{\mathrm{ADM}}|\, \epsilon \oint (\cos\theta_0, \sin\theta_0)\, d\theta_0 = 0, \tag{17}$$

the warp-bubble analogue of Purcell's scallop theorem: a reciprocal deformation cycle yields no translation. Any directional bias must be *manufactured* by breaking the reciprocity of the swing.

#	Domain	Source	Catcher verdict / location
1	D-CTC	`phase_a`	iter regime change at $\dim_{\mathrm{CR}} \geq 6, \dim_{\mathrm{CTC}} \geq 4$
2	D-CTC	`phase_oqr`	Haar vs Clifford eigenvalue divergence
3	D-CTC	`phase_e`	purity-quantile $\sigma_{\min}$ threshold
4	D-CTC	`phase_f`	purity-quantile $\sigma_{\min}^{\mathrm{joint}}$ threshold
5	D-CTC	`phase_g`	purity-quantile best-overlap threshold
6	D-CTC	`E4_trichotomy`	basin boundaries (rows $1 \to 2$ and $8 \to 9$)
7	HW	Marrakesh batch 5	extended extinction zone $\delta \in [7\pi/8, 3\pi/2]$
8	Phase module	`cauchy_stability`	sharp at $r = 2.28$ ($27\times$ median)
9	Phase module	`spinor_monodromy`	sharp at $r = 1.67$ ($14\times$)
10	Phase module	`synchrotron_analog`	sharps at $r = 1.36, 3.52, 7.84$
11	Phase module	`kg_scattering`	sharp at $r = 1.976$ ($16\times$)
12	Phase module	`photon_sphere`	sharp at $r = 1.976$ ($32\times$); **coincides with #**
13	Phase module	`berry_phase_lp`	sharp at $r = 5.58$ ($13\times$)
14	Cover/cascade	`stress_zn_closure`	transitions across $N \in \{2, 3, 5, 6, 7, 11, 17\}$
15	Warp hybrid	`tipler_krasnikov_hybrid`	CTC-band on/off NEC gating
16	Phase module	`optical_fiber_analog`	analog horizon onset at v_{end}=0.669
17	Feedback shell	`feedback_amplified_shell`	critical $Q \approx 7.86$ threshold
18	Composite	`knopp_drive`	sharp at $r = 12$ (band re-entry)
19	Warp ring	`krasnikov_ring`	N=2 pair fragile to phase noise at $\sigma \approx 2.058$ r
20	HW	Marrakesh batch 6	Knopp Drive band gating HW-confirmed on `i`
21	HW	Kingston batch 7	Band edges $r_{\mathrm{in}} = 2.657 \pm 0.002$, $r_{\mathrm{out}} = 5.471 \pm$
22	HW cross-validation	`knopp_derivative_catcher_validation`	Derivative catcher independently identifies Ki
23	HW cross-chip	`knopp_cross_chip`	Kingston + Marrakesh batch 7 agree at RMS=

Table 3: The twenty-three Knopp-Drive-relevant catcher-verified emergents. Items 1–6 are from the Deutsch-CTC family. Items 7 and 20 are hardware-validated on the IBM Quantum `ibm_marrakesh` 156-qubit Heron-r2 processor. Items 8–13 are radial sharps in the Systrophē LP supercritical exterior; the coincidence of #11 and #12 at $r = 1.976$ is partially explained by their shared F/L dependence (not truly independent diagnostics). Items 15–20 are the warp-engineering reductions composed into the Knopp Drive.

12.2 The rising-floor ratchet

We break reciprocity with a port of the Pareto ratchet proven in the TriCameral control stack. Each pendulum cycle is a *power stroke* ($\epsilon = \epsilon_{\mathrm{hi}}$, worldline outside the band, exotic matter spent and steering thrust produced) followed by a *recovery stroke* routed through the Tipler band, where the gate factor $\to 0$ makes the return free. A two-metric ratchet on (displacement d, pinch-safety $s = 1 - \epsilon_{\mathrm{hi}}$) accepts a stroke only when the product $d \cdot \max(s, \delta)$ improves and, crucially, raises its acceptance floor monotonically,

$$\text{floor} \leftarrow \max\!\big(\text{floor},\ \text{best} \times f\big), \qquad f \in (0, 1), \tag{18}$$

so the configuration can never relax below f of its best lean; a stroke regressing below the raised floor rolls back to the saved best state, giving closed-loop stability rather than open-loop runaway. The measured cost asymmetry between advancing and retreating one ratchet step, sourced entirely from `knopp_budget`, rises with f from $1.7\times$ (no pawl; the residual gate asymmetry alone) through $7.3\times$ ($f = 0.85$) to $> 100\times$ ($f = 0.99$). The bias is reversible: re-anchoring with $\theta_0 \to \theta_0 + \pi$ locks the opposite direction.

12.3 The shell as a surface capacitor

The Q-cavity of Section 3.3 acts as an energy-accumulating surface capacitor. Under the parametric (dynamical-Casimir) pump it rings up as $E_{\mathrm{shell}}(t) = E_{\mathrm{shell}}(0)e^{gt}$ and at saturation holds only

$$E_{\rm peak} = |E^{\rm bare}_{\rm neg}|/Q \tag{19}$$

sustained over the cavity lifetime $\tau = Q/f_0$. The instantaneous negative-energy charge present on the bubble is thus reduced by a factor Q and the sustained pump power by $1/Q^2$: one never marshals the full requirement at once, but trickle-charges a small standing charge. The negative energy itself is the parametrically squeezed Casimir vacuum (the dynamical Casimir effect [15]), *not* a reservoir of exotic matter; static parallel-plate Casimir is quantitatively insufficient, and only the amplified route reaches the budget while respecting the quantum inequality.

12.4 Ford–Roman quantum-interest ledger

Negative energy borrowed for a hold time τ must be repaid by a larger positive pulse (the Ford–Roman quantum-interest conjecture [14]). Modelling the premium as $r = (\tau/\tau_0)^\alpha - 1 = Q^\alpha - 1$ with pulse width $\tau_0 = 1/f_0$, the per-stroke repayment at $\alpha = 1$ is

$$E_{\rm repay} = \frac{|E^{\rm bare}_{\rm neg}|}{Q}(1 + r) = |E^{\rm bare}_{\rm neg}|, \tag{20}$$

independent of Q: amplification removes the pump overhead $1/Q$ but never the quantum-interest *principal*. The total per-stroke ledger $E = E_{\rm pump} + E_{\rm repay} = |E^{\rm bare}_{\rm neg}|(1 + 1/Q)$ therefore falls toward an irreducible floor as $Q \to \infty$ (within +0.1% at $Q = 10^3$). For superlinear interest $\alpha > 1$ the cost per unit displacement instead has an interior optimum at $Q^* = (\alpha - 1)^{-1/\alpha}$.

12.5 SI feasibility floor

Converting the irreducible principal to SI via the textbook geometrised estimate $E_{\rm geom} \simeq \kappa(v_s/c)^2 R^2/\sigma$ ($\kappa \approx 1/12$) and $c^4/G = 1.21 \times 10^{44}$ J/m, a one-metre luminal bubble ($\sigma = R$) carries a floor of $\approx 6.5 \times 10^{44}$ J ≈ 3.8 Jupiter mass-energies, scaling as R^2/σ. This exceeds the negative-energy budget of a state-of-the-art squeezed-vacuum source by ~ 60 orders of magnitude. The ratchet and capacitor make the *peak charge* and *pump power* arbitrarily small, but this *total* is the same irreducible wall every warp metric encounters. The full controller is implemented in the `systrophe.knopp_ratchet` module, with the address-space catcher returning `smooth` on the bias-energy curve (a continuously tunable control, not an emergent transition).

13 The irreducible energy floor, the wormhole map, and the buildable residue

The bias controller (Section 12) lowers the *peak* negative-energy charge and pump power but not the time-integrated Ford–Roman principal. We close, from first principles, the question of whether *any* route lowers that principal, and identify the one capability that is genuinely buildable.

13.1 Geometry does not lower the principal

The principal scales as $E \sim (c^4/G)\,(v_s/c)^2 R^2/\sigma$, so the Van den Broeck [7] trick—decoupling the energy-driving shell radius from the passenger pocket—appears to help. Calibrating the neck blow-up against the exact Einstein-tensor integral of the conformally-flat static pocket, $E_{\rm pocket} = \frac{1}{2}\int (B'^2/B) r^2\,dr$, the energy scales as $B_{\max}$ (not $\ln^2 B_{\max}$), so the blow-up dominates the shell and re-inflates the floor to $\sim (c^4/G)R_{\rm pocket}$. A calculus-of-variations argument (Euler–Lagrange minimiser $B = (1 + c/r)^2$, isotropic Schwarzschild) shows the thickest wall is optimal yet still gives $E_{\min} \approx 2(c^4/G)\rho_{\rm use}$; the direct Einstein-tensor integral of a *regular* pocket gives an exotic (NEC-violating) energy $\approx 5(c^4/G)\rho_{\rm use}$. The interior stays flat and habitable, but the floor is Jupiter-scale and irreducible across all wall shapes.

13.2 The wormhole map: energy wall = entanglement wall

Mapping the traversable-wormhole routes, the classical Morris–Thorne exotic energy and the quantum ER=EPR [3] requirement are dual: a throat of radius r_t needs either $\sim (c^4/G)r_t$ of exotic energy *or* $\sim (r_t/\ell_P)^2$ ebits of pre-shared entanglement (Ryu–Takayanagi $S = A/4G_N$). The classical exotic energy equals exactly half the ER=EPR dual black-hole mass-energy at every scale—both equal assembling a black hole of radius r_t. For matter transport the wall is the same Bekenstein bound; only *information* traverses cheaply.

13.3 Hardware-validated ER=EPR channel

The information channel is buildable. Under ER=EPR the shared entanglement is the bridge; sending a qubit through it (Gao–Jafferis–Wall [4] teleportation) is the operational traversal. We verify, from first principles, that the Maldacena–Qi coupling $V = \frac{1}{n}\sum_j Z_j^L Z_j^R$ is the operator-size operator on the EPR state, $V\, P_L|\mathrm{EPR}\rangle = (1 - 2\,\mathrm{size}(P)/n)\, P_L|\mathrm{EPR}\rangle$ (residual 10^{-17}); a real SYK scrambler activates the size-winding channel where a Haar scrambler is inert, but the faithful dense-SYK wormhole transpiles to depth 11570 (3816 two-qubit gates) and cannot run clean on current hardware. The deterministic coherent channel does: executed on IBM Heron-r2 hardware with a live-calibration best-qubit pipeline and error mitigation, the entanglement-teleportation Bell fidelity reaches **0.958** (`ibm_kingston`, zero-noise extrapolated; 0.918 on `ibm_marrakesh`), well above the 0.5 classical bound. This is entanglement-mediated quantum information transport—a real quantum-network primitive—*not* faster-than-light and *not* matter transport.

14 Monetisation pathway

The Knopp Drive composite provides four independently-quantified engineering bounds, each directly relevant to warp-drive physical realisation efforts:

1. **Geometric exotic-matter zero**: a Krasnikov-tube craft routed through a Tipler CTC band requires *zero* exotic matter for as long as the worldline lies inside the band. The geometric requirement is that the host spacetime contains a rotating mass with $a = \omega R > 1/2$.

2. **Power-budget law**: feedback-amplified warp shells scale sustained drive power as $1/Q^2$ in the high-Q limit, with a hard transition at $Q \approx 7.86$ (catcher emergent #17) below which feedback is ineffective.

3. **Steering law**: continuous steering at $|\boldsymbol{p}| = R^2\epsilon|m_{\mathrm{ADM}}|$ up to $\epsilon \to 1$, with no threshold-locked discontinuity.

4. **Pfenning–Ford fence**: the composite saturates but never violates the quantum inequality.

These four bounds are jointly novel and are reduced to an executable Python module (`systrophe.knopp_drive`) with full numerical implementation, tests, and an example walk-through.

15 Toroidal Knopp Drive: finite-length realisation via a counter-rotating Kerr binary

The idealised Tipler source — an infinite, rigid, axisymmetric dust column — is the single largest obstruction to physical realisability of the Knopp Drive composite. Aguilera Katayama (April 2026, DOI 10.13140/RG.2.2.16820.82568) recently proposed a finite-length effective realisation in which the infinite cylinder is replaced by the inter-horizon toroidal domain of a gravitationally bound binary of two near-extremal Kerr black holes with maximally antiparallel (counter-rotating) spins. The composite Lense–Thirring frame-dragging in the equatorial toroidal domain is diffeomorphic — in a coarse-grained effective sense — to the Bonnor Case III exterior's CTC bands. This section integrates that construction into the Knopp Drive stack and reports a self-consistent reduced engineering budget.

15.1 Effective backend: `EffectiveToroidalKerrBinary`

For two equal-mass Kerr holes with masses $M_1 = M_2 = M$, dimensionless spin $\chi \approx 1$, antiparallel along the binary axis, and coordinate separation d, the leading-order Lense–Thirring angular velocity in the midplane ($z = 0$) at cylindrical radius ρ from the axis is, by linear superposition,

$$\Omega_{\rm eff}(\rho) \;=\; \sum_{i=1,2} \frac{2\,a_i M_i}{r_i^3} \;\approx\; \frac{4\chi M^2}{[(d/2)^2+\rho^2]^{3/2}} \qquad (G=c=1), \tag{21}$$

with $r_i = \sqrt{(d/2)^2+\rho^2+z^2}$. The two contributions add constructively because the spins are antiparallel and the field is gravitomagnetic. The effective Tipler tilt — the binary analogue of $T(r) = |K(r)|/|F(r)|$ in the Bonnor Case III exterior — is

$$T_{\rm eff}(\rho) \;\equiv\; \frac{|g^{\rm eff}_{t\phi}|}{|g^{\rm eff}_{tt}|} \;\approx\; \frac{\Omega_{\rm eff}(\rho)\,\rho^2}{1-2\Phi_{\rm eff}(\rho)} \;\xrightarrow[\text{outer region}]{}\; \frac{4\chi M^2\rho^2}{[(d/2)^2+\rho^2]^{3/2}}. \tag{22}$$

The Tipler gate of §3.1 carries over unchanged: $g^{\rm eff}_{\rm Tipler}(\rho) = \max(1 - c\,T_{\rm eff}(\rho), 0)$.

15.2 Toroidal CTC band: existence threshold

Setting $T_{\rm eff}(\rho) = 1$ and substituting $x = \rho/(d/2)$, $k = 2M/d$ (we set $\chi = 1$):

$$(2kx)^2 \;=\; (1+x^2)^{3/2}. \tag{23}$$

The minimum of the right-hand side over the left-hand side, $\min_{x>0}(1+x^2)^{3/2}/x^2$, occurs at $x^2 = 2$ and equals $3\sqrt{3}/2 \approx 2.598$. The band-existence condition is therefore

$$\boxed{k \;=\; \frac{2M}{d} \;\geq\; k_{\rm crit} \;=\; \sqrt{\frac{3\sqrt{3}}{8}} \;\approx\; 0.806.} \tag{24}$$

A toroidal CTC band exists *only* when $d \lesssim 2.48\,M$. The Aguilera Katayama paper's quoted reference configuration ($M = 10^6 M_\odot$, $d = 10M$, $k = 0.2$) lies *below* this threshold and does *not* admit a band in the leading-order LT approximation; the working configuration requires tight ($k \gtrsim 0.806$), maximally counter-rotating binaries.

15.3 Composite budget in the toroidal regime

Inside the toroidal CTC band ($T_{\rm eff} \geq 1$, $g^{\rm eff}_{\rm Tipler} = 0$), the master composite formula of §4 survives with $T(r) \to T_{\rm eff}(\rho)$:

$$\left|E^{\text{Knopp-Toroidal}}_{\rm neg}\right| \;=\; \left|E_{\rm Krasnikov}(\alpha_{\rm wall},\sigma)\right| \cdot g^{\rm eff}_{\rm Tipler}(\rho)\cdot\frac{1}{Q^2}\cdot(1+\epsilon). \tag{25}$$

Classically inside the band, $|E^{\text{Knopp-Toroidal}}_{\rm neg}| = 0$. The chronology-protection counter-flux from semiclassical back-reaction is estimated as

$$E_{\rm BR} \;\sim\; \lambda\cdot\frac{1}{M^4}\cdot\frac{\rho^2 d}{2880\pi^2}, \tag{26}$$

with λ a small $\mathbb{Z}_3$-cover prefactor (default 10^{-4}). The Newton–Kantorovich loop (§6) is expected to converge to a self-consistent fixed point in which the back-reaction flux is absorbed by the Q-cavity once $Q \gtrsim Q_{\rm thr} = \sqrt{|E_{\rm Krasnikov}|/E_{\rm BR}}$. For the working configuration below this threshold is at the level of 10^2–10^3.

15.4 Working configuration

A representative configuration where $g^{\text{eff}}_{\text{Tipler}} = 0$ classically (tight, maximally counter-rotating binary in band):

parameter	value	meaning
M	1 (geometric units)	individual BH mass
d	$2M$	axial separation ($k = 1 > k_{\text{crit}}$)
χ	1	near-extremal antiparallel spin
ρ_{orbit}	$1.5\,M$	orbit radius inside band
Q	100	cavity quality factor
ϵ	0.01	horn-twist amplitude
band edges	$[0.65,\, 3.57]\,M$	numerically solved
$T_{\text{eff}}(1.5M)$	1.54	inside band
$g^{\text{eff}}_{\text{Tipler}}$	0	classical zero
composite $\lvert E_{\text{neg}}\rvert$	0	classical
back-reaction E_{BR}	1.6×10^{-8}	semiclassical residual

The full toroidal Knopp budget is implemented in `systrophe.knopp.knopp_toroidal` (`EffectiveToroidal`, `knopp_toroidal_budget`). The IBM hardware-validation circuits of §7 carry over unchanged: the Floquet eigenmode encoding maps to the toroidal-band frequencies via the diffeomorphism, and the band-edge extinction signature is the same up to a frequency rescaling.

15.5 Honest caveats specific to the toroidal extension

- The Lense–Thirring superposition is linear; corrections from the near-horizon strong field can shift T_{eff} by $O(1)$ via the $(1 - 2\Phi_{\text{eff}})$ denominator, and may eliminate the band entirely when included.
- Stability of a near-extremal antiparallel-spin binary against orbital decay and merger is itself a speculative claim of the framework. The working configuration above is a classical kinematic statement, not a dynamical viability result.
- The diffeomorphic isomorphism to Bonnor Case III is an effective-correspondence claim, not a proven global isometry. The exact composite spacetime is not in closed form.

15.6 Self-consistent semiclassical back-reaction via Newton–Kantorovich

The Aguilera Katayama framework leaves the back-reaction term as a dimensional ansatz $E_{\text{BR}} \sim \lambda\, M^{-4}\, \rho^2 d/(2880\pi^2)$ with a free prefactor λ. We replace it here by a Newton–Kantorovich iteration of the band-edge equation under an actual Polyakov–Boulware vacuum source.

On the 2D radial slice $ds^2_{\text{2D}} = -F(\rho)\,dt^2 + d\rho^2$ with $F(\rho) = 1 - 2\Phi_{\text{eff}}(\rho) = 1 + 4M/\sqrt{(d/2)^2 + \rho^2}$, the Boulware vacuum stress

$$\langle T_{tt}\rangle_B(\rho) \;=\; -\frac{1}{24\pi}\left[F''(\rho) - \frac{F'(\rho)^2}{2F(\rho)}\right] \tag{27}$$

is the exact Davies–Fulling–Unruh result in the conformally-flat 2D sector. We define the back-reacted Tipler tilt

$$T^{\text{BR}}_{\text{eff}}(\rho;\lambda) \;=\; T^{\text{classical}}_{\text{eff}}(\rho) \;-\; \lambda\,\langle T_{tt}\rangle_B(\rho) \tag{28}$$

and solve the band-edge condition $T^{\text{BR}}_{\text{eff}}(\rho;\lambda) = 1$ via `newton_kantorovich_1d` for each edge. Bisection over λ identifies λ_{crit} – the back-reaction strength above which the band collapses.

For the working configuration ($M{=}1$, $d{=}2M$, $k{=}1$):

- Classical band: $\rho \in [0.6523,\, 3.5719]\, M$.
- $\langle T_{tt} \rangle_B$ is sign-changing: $+6.8\times10^{-3}$ at the inner edge, -1.6×10^{-3} at the outer.
- Newton–Kantorovich on each edge converges in $\lesssim 20$ iterations.
- $\lambda_{\rm crit} \approx 21.58$ – the band closes for $\lambda \geq \lambda_{\rm crit}$.

The Q-cavity threshold follows from energy balance between the chronology-protection power flux and the cavity absorption rate:

$$\boxed{Q_{\rm thr} \;=\; \sqrt{\frac{|E_{\rm Krasnikov}|\,\omega_0\,A_{\rm band}}{|\langle T_{tt}\rangle_B|_{\rm inner}}}} \tag{29}$$

For the working configuration with $|E_{\rm Krasnikov}| = \omega_0 = 1$, this evaluates to $Q_{\rm thr} \approx 75.3$ – within 13% of the dimensional estimate "$Q \gtrsim 86$" in the Aguilera Katayama framework, and now anchored in an explicit Polyakov vacuum stress rather than a free constant. The full implementation is `systrophe.knopp.knopp_toroidal_backreaction` (`backreaction_diagnostic`, `critical_lambda`, `q_threshold_from_balance`).

15.7 Dynamical stability and gravitational-wave signature

The framework's most speculative claim is that two near-extremal counter-rotating Kerr black holes hold together long enough to host the toroidal CTC band. We quantify this with the classical-GR (Peters 1964 quadrupole + leading-order Kidder 1995 spin–spin) result.

Spin–spin coupling at the working point. For maximal antiparallel spins $\boldsymbol{S}_1 = +M^2\hat{n}$, $\boldsymbol{S}_2 = -M^2\hat{n}$ aligned with the binary axis,

$$U_{\rm SS} \;=\; \frac{1}{r^3}[3(\boldsymbol{S}_1\cdot\hat{n})(\boldsymbol{S}_2\cdot\hat{n}) - \boldsymbol{S}_1\cdot\boldsymbol{S}_2] \;=\; -\frac{2\chi^2 M^4}{r^3}. \tag{30}$$

The ratio to the Newtonian orbital energy $E_{\rm orb} = -M^2/(2d)$ is $|U_{\rm SS}/E_{\rm orb}| = 4\chi^2 M^2/d^2$. For the working configuration ($M=1$, $d=2M$, $\chi=1$) this is **1.00** – the binary is fully non-perturbative for spin–spin.

Inspiral lifetime. The Peters merger time for a circular equal-mass binary is $t_{\rm merge} = (5/256)\, d^4/M^3$ (geometric units). For the working configuration this gives $t_{\rm merge} = 0.3125\, M$, while the orbital period is $T_{\rm orb} = 2\pi/\Omega_{\rm orb} = 2\pi\sqrt{2}\, M^{1/2} = 8.886\, M$, so the inspiral lasts only

$$\boxed{N_{\rm orbits} \;=\; t_{\rm merge}/T_{\rm orb} \;\approx\; 0.012} \tag{31}$$

– the binary merges in about 1% of a single orbit. The toroidal CTC band, which requires the binary to persist, cannot persist either.

GW signature. The dominant quadrupole emission peaks at $f_{\rm GW} = \Omega_{\rm orb}/\pi \approx 0.159/M$, which in SI units (with M in solar masses) is $f_{\rm GW} \approx 3.23\times10^4/M_\odot$ Hz. For an $M = 10\, M_\odot$ stellar binary this lands in the LIGO band (~ 3.2 kHz); for $M = 10^6\, M_\odot$ in the LISA band ($\sim 3.2\times10^{-2}$ Hz). The merger time in seconds for $M = 10\, M_\odot$ is $\sim 7.7\,\mu$s.

Verdict. The classical-GR analysis falsifies the working configuration of the Toroidal Knopp Drive on dynamical grounds. The two regimes available are:

- $d \lesssim 2.48\,M$ (band exists, $k \geq k_{\mathrm{crit}}$): binary survives <1 orbit. NOT VIABLE.
- $d > 2.48\,M$ (binary survives many orbits): NO toroidal CTC band. ALSO NOT VIABLE.

Any rescue of the toroidal Knopp Drive must invoke (a) non-gravitational binding (an unmodelled force or composite structure that holds the binary tight against quadrupole decay), or (b) a beyond-leading-order GR effect that strongly suppresses inspiral in the extremal counter-rotating regime, or (c) a different binary topology (e.g., a non-equal-mass or non-aligned configuration). The full classical analysis is implemented in `systrophe.knopp.knopp_toroidal_stability` (`stability_report`, `corrected_merger_time`, `gw_frequency_in_hz`, `detector_classification`).

15.8 Rescue paths explored

The three rescue paths flagged in §15.7 – non-gravitational binding, beyond-leading-order GR, and alternative binary topologies – are tested concretely in `systrophe.knopp.knopp_toroidal_pn_rescue`, `knopp_toroidal_asymmetric`, and `knopp_necklace`. The standing verdicts:

(a) PN rescue. At $d = 2M$ ($v^2 = M/r = 0.5$), the 1PN luminosity correction coefficient $-3.71 - 0.73\nu \approx -3.89$ at $\nu = 1/4$ flips the sign of the PN factor $1 + a_1 v^2 \approx -0.95$. The PN series has broken down in the strong-field regime; no reliable PN truncation can be quoted. At larger separations $d \gtrsim 5M$ the PN series is well-behaved but no CTC band exists there (k < k$_c rit$).**Rescue path #2 is closed.**

(b) Asymmetric topology. A scan of $\{q, \alpha, d\}$ at $q \in \{1, 0.5, 0.1, 0.01\}$, $\alpha \in \{\pi, 0.9\pi, 0.7\pi, 0.5\pi\}$, $d \in \{1.5, 2, 2.5, 5, 10\}\,M$ (80 configurations) finds zero parameter points where the toroidal CTC band exists *and* the binary survives a single orbit. Unequal-mass binaries gain inspiral lifetime (chirp mass shrinks as $q^{3/5}$) but lose CTC band coverage (kappa-factor degradation under misalignment, plus the band-existence threshold moves). **Rescue path #3 is closed.**

(c) Multi-binary necklace. A ring of N counter-rotating Kerr binaries at angular positions $\phi_k = 2\pi k/N$ extends the band geometrically: the necklace closes into a continuous CTC band along the ring once $N \geq N_{\mathrm{pack}}$, where N_{pack} depends on R_{ring}/d. For $R_{\mathrm{ring}} = 5M$, $d = 2M$: $N_{\mathrm{pack}} = 4$. However, each individual binary inherits the same Peters merger problem ($n_{\mathrm{orbits}} \approx 0.025$ per binary). The necklace does *not* fix the inspiral; only mutual angular-momentum exchange or ring-mode tidal coupling could, and neither is in the leading-order construction. **The necklace extends band coverage geometrically but inherits the falsification.**

Net. None of the three rescue paths flagged in §15.7 closes the inspiral-vs-band gap under the classical-GR + leading-order PN analysis. Any rescue must come from a yet-unmodelled effect: non-gravitational binding force, full numerical-relativity extremal-spin suppression, or a fundamentally different binary topology. Each is documented in the corresponding module's docstring but not derived here.

15.9 Quantum-gravity probes of the toroidal band

Two probes from the broader Systrophe stack become well-defined when the CTC band is the *finite* toroidal region rather than the infinite Tipler cylinder (where both diverge):

LQG area discretization. The band's inner and outer boundaries are 2-tori of proper area $A = 2\pi\rho L_z\sqrt{1-2\Phi_{\mathrm{eff}}(\rho)}$. The LQG quantization $A_j = 8\pi\gamma\ell_P^2\sqrt{j(j+1)}$ (Immirzi $\gamma \approx 0.2375$) gives a half-integer spin j for each boundary, with a discretization residual that scales as $A_{\mathrm{classical}}^{-1/2}$. For the working configuration ($M = 1$, $d = 2M$, $\rho \in [0.65, 3.57]M$), $j_{\mathrm{inner}} \approx 2.5$, $j_{\mathrm{outer}} \approx 10.5$, with residuals of 3.3% and 1.4% respectively. The band width in Planck units is $\Delta\rho \approx 2.9\,\ell_P$ – well above unity, so the band is resolvable by the spinfoam; the band would be *protected by gravitational discreteness alone* if its width fell below the spin-network coarse-grain scale.

Holographic complexity (CV & CA). The toroidal volume $V_{\mathrm{band}} = \int_{\rho_{\mathrm{in}}}^{\rho_{\mathrm{out}}} 2\pi\rho\sqrt{1-2\Phi}\,d\rho\cdot L_z$ gives a Complexity = Volume proxy

$$\mathcal{C}_V \;=\; \frac{V_{\mathrm{band}}}{G_N\,\ell_{\mathrm{AdS}}}, \qquad \ell_{\mathrm{AdS}} \equiv d. \tag{32}$$

A Complexity = Action proxy uses the gravitomagnetic-energy density:

$$\mathcal{C}_A \;\sim\; \frac{1}{\pi}\int_{V_{\mathrm{band}}} \Omega_{\mathrm{eff}}^2\,dV. \tag{33}$$

The Lloyd bound caps the complexity-growth rate at $d\mathcal{C}/dt \le 2E/(\pi\hbar)$ with $E = 2M$. For the working configuration this gives $d\mathcal{C}/dt_{\max} \approx 1.27$ and a CV-traversal time $\tau_V = \mathcal{C}_V/\dot{\mathcal{C}}_{\max} \approx 49\,M$. Both proxies vanish as $d/M \to 1/k_{\mathrm{crit}}$, where the band closes.

The full quantum-probe module is `systrophe.knopp.knopp_toroidal_quantum`, with combined diagnostic `toroidal_quantum_diagnostics(binary)`.

16 Limitations and falsifiers

Linearised composition. The four mechanisms are composed in the linearised (single-Krasnikov-tube perturbation of LP exterior) regime. Cross-products of perturbations are formally $O(G^2)$ and not modelled.

Idealised Tipler source. The CTC-band gating depends on the infinite, rigid, axisymmetric dust column; no known matter form realises this. Finite-length corrections are explored in §15 via the counter-rotating Kerr binary toroidal realisation, which is buildable in principle but inherits its own speculative stability claims.

Quantum-inequality saturation. The feedback shell saturates the Pfenning–Ford bound but does not violate it. Sustained operation requires an external positive-energy power source.

Horn pinch. The horn-twist parameter ϵ must satisfy $\epsilon < 1$ to avoid horn-torus self-intersection. The linear $\boldsymbol{p} \sim \epsilon$ scaling is reliable on $[0, 0.9]$.

Irreducible energy floor. The reversible-ratcheting bias controller (Section 12) lowers the *peak* negative-energy charge and the pump power without bound as $Q \to \infty$, but the Ford–Roman quantum-interest principal is invariant and sits at Jupiter-mass scale ($\sim 10^{44}$–10^{45} J for a metre-scale bubble), ~ 60 orders of magnitude beyond laboratory squeezed-vacuum sources. The controller is an engineering convenience over an irreducible energy wall, not a route around it.

Chronology protection. The Knopp Drive is pre-quantum: the chronology-protection conjecture [11] is not engaged. The Deutsch-CTC analysis [12] in the Systrophē DCTC trilogy provides one route to a quantum-consistent CTC theory.

Numerical validation only. All four mechanisms are validated by Python simulation and the address-space novelty catcher; the composite has not been demonstrated on quantum hardware. Marrakesh batch 5 verified the bubble-extinction property at the wall level on IBM `ibm_marrakesh` (5/5 circuits, TV < 0.05); the Knopp composite has not yet been mapped to a hardware experiment.

17 Conclusion

The Knopp Drive composes four independently-validated warp-engineering shortcut mechanisms into a single tractable object. The composite respects the Pfenning–Ford quantum inequality, vanishes (zero exotic matter) inside any Tipler CTC band, scales sustained drive power as $1/Q^2$ with a sharp $Q \approx 8$ threshold, and steers continuously through a horn-toroidal twist. A reversible-ratcheting pendulum bias controller (Section 12) converts the direction-blind steering dipole into stable directed transport and books the journey against the Ford–Roman quantum-interest bound, which fixes an irreducible Jupiter-mass energy floor that amplification reduces in *peak* and *power* but cannot remove in *total*. We further establish, from first principles (Section 13), that *no* route lowers that principal: Van den Broeck geometry re-inflates under exact Einstein-tensor calibration, the thick-wall variational optimum still costs $\sim (c^4/G)\rho_{\text{use}}$, and the wormhole energy wall is dual to a Bekenstein entanglement wall. The honest residue is not matter transport but an *information* channel: the ER=EPR teleportation primitive, hardware-validated on IBM Heron-r2 at 0.958 Bell fidelity—a genuine quantum-network result, not a propulsion device. The full implementation lives in the `systrophe.knopp_drive`, `knopp_ratchet`, `wormhole_map`, `size_winding`, and `erepr_channel` modules of the open-source *Systrophē* package [13].

Acknowledgements

This work is part of the larger Systrophē framework. Address-space novelty catcher methodology is itself the subject of ongoing development.

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